



2 Real Analysis 2023

Tutorial 12

A/Prof Elena Berdysheva
Dr Francesco Russo



Uniform continuity

Problem 51 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = \begin{cases} -1, & x \in (-\infty, 0), \\ \frac{1}{2-x}, & x \in [0, 2), \\ 0, & x \in [2, \infty). \end{cases}$$

Prove the following statements. You may prove them by definition or using known statements from the lecture.

- (a) f is not continuous on the interval $[-1, 1]$,
- (b) f is uniformly continuous on the interval $[0, \frac{3}{2}]$,
- (c) f is continuous, but not uniformly continuous on the interval $(0, 2)$.

Problem 52 (a) Use the Mean Value Theorem to prove that $|\sin x - \sin y| \leq |x - y|$ for any $x, y \in \mathbb{R}$.

- (b) Show that the functions $f(x) = x$ and $g(x) = \sin x$ are both uniformly continuous on $\mathbb{R}$.
- (c) Show that the function $h(x) = x \sin x$ is not uniformly continuous on $\mathbb{R}$.

Uniform convergence

Problem 53 Consider the sequence of functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$, $f_n(x) = x^n(1 - x)$, $n \in \mathbb{N}$.

- (a) Determine the pointwise limit f of the sequence $(f_n)_{n=1}^\infty$ on the interval $[0, 1]$.
- (b) For each $n \in \mathbb{N}$, find the maximum value of f_n on $[0, 1]$.
- (c) Does $(f_n)_{n=1}^\infty$ converge to f uniformly on $[0, 1]$?

Problem 54 Consider the sequence of functions $g_n : [0, \infty) \rightarrow \mathbb{R}$, $g_n(x) = \frac{nx}{1+nx}$, $n \in \mathbb{N}$.

- (a) Determine the pointwise limit g of the sequence $(g_n)_{n=1}^\infty$ on $[0, \infty)$.

- (b) Show that $(g_n)_{n=1}^{\infty}$ converges to g uniformly on $[1, \infty)$.
- (c) Show that $(g_n)_{n=1}^{\infty}$ does not converge to g uniformly on $[0, \infty)$.

Problem 55 Consider the sequence of functions $h_n : \mathbb{R} \rightarrow \mathbb{R}$, $h_n(x) = e^{-nx}$, $n \in \mathbb{N}$.

- (a) Determine the pointwise limit h of the sequence $(h_n)_{n=1}^{\infty}$ on $[0, \infty)$.
- (b) Does $(h_n)_{n=1}^{\infty}$ converge to h uniformly on $[0, 1]$?

Series of functions

Problem 56 Show that the series $\sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$ converges absolutely and uniformly on $\mathbb{R}$.

Problem 57 Show that for any $r > 0$ the series $\sum_{n=1}^{\infty} \frac{1}{x^2n^2}$ converges absolutely and uniformly on the set $\{x \in \mathbb{R} : |x| \geq r\}$.

Problem 58 Show that for any $r > 0$ the series $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ converges absolutely and uniformly on the interval $[-r, r]$.

Problem 59 Show that the series $\sum_{n=0}^{\infty} \frac{1}{1+x^n}$ does not converge for any $x \in (-1, 1]$, but converges on $(1, \infty)$.

Problem 60 Consider the series $\sum_{n=0}^{\infty} \frac{x^{2n}}{1+x^{2n}}$. Prove or disprove:

- (a) The series converges on $[-1, 1]$.
- (b) The series converges on $(-1, 1)$.
- (c) The series converges on $[2, 3]$.
- (d) The set of points where the series converges is $(-1, 1)$.
- (e) For all values x for which the series converges we have $\sum_{n=0}^{\infty} \frac{x^{2n}}{1+x^{2n}} \geq \frac{1}{2}$.

Power series

Problem 61 For each of the following series, determine its radius of convergence.

$$(a) \sum_{n=0}^{\infty} \frac{x^n}{n^n}, \quad (b) \sum_{n=0}^{\infty} \frac{n^n x^n}{n!}, \quad (c) \sum_{n=0}^{\infty} \frac{(n!)^2 x^n}{(2n)!}.$$